

- Normal python lists are general-purpose.
- Numpy arrays are optimized for numerical computation.

→ import numpy as np

→ #create

a = np.array([1, 2, 3, 4])

12. Numpy array vs Python list

py list : $a = [1, 2, 3]$

numpy : $a = \text{np.array}([1, 2, 3])$

• differences $\rightarrow$

Python list

1. heterogeneous possible
2. general purpose
3. slower num loops

$a = \text{np.array}([1, 2, 3])$

$\text{print}(a * 2)$

o/p - $[2, 4, 6]$

Numpy array

1. homogenous num storage
2. vectorized operations
3. optimized low-level impl

$[1, 2, 3] * 2$

o/p - $[1, 2, 3, 1, 2, 3]$

13. Shape

$X = \text{np.array}([$
 $[1, 2, 3],$
 $[4, 5, 6]$
 $])$

Shape : $X.\text{shape}$
returns $(2, 3)$

In ML -

$X.\text{shape} = (\text{samples}, \text{features})$

example:

$(1000, 20)$

means 1000 samples

20 features each.

14. Dimensions

$X.\text{ndim}$

$\rightarrow \text{np.array}([1, 2, 3])$

shape: $(3,)$ that gives 1D.

for images - (height, width, channels)

example : $224 \times 224 \times 3$

15. Indexing

$X = \text{np.array}([$
 $[10, 20, 30],$
 $[40, 50, 60],$
 $[70, 80, 90]$
 $])$

$\Rightarrow \text{array}[\text{row}, \text{column}]$

get 1st row :

$X[0]$

get 1st row, 2nd col :

$X[0, 1]$

column :

$X[:, 1]$

gives $[20, 50, 80]$

16. Reshaping

$a = \text{np.array}([1, 2, 3, 4, 5, 6])$

reshape :

$a.\text{reshape}(2, 3)$

result :

1 2 3
4 5 6

shape :

$(2, 3)$

Rule : number of elements must be equal

6 elements can become :-

1×6

2×3

3×2

6×1

but not :

4×2 + requires 8 values

17. -1 in reshape

a.reshape(-1, 2)

Numpy calculates the missing dimension.

18. Axis ***

```
X = np.array([
    [1, 2, 3],
    [4, 5, 6]
])
```

X.sum(axis=0)

explanation =
axis = 0

↓
collapses vertically
↓
operate column-wise

```
1 2 3
4 5 6
↓ ↓ ↓
[5 7 9] ≠ result
```

→ X.sum(axis=1)

explanation =
axis = 1

↓
collapse horizontally
↓
operate row-wise

```
1 2 3 → 6
4 5 6 → 15
[6, 15] → result
```

19. Vectorization ** In

Normal loop:

result = []

for x in nums

result.append(x*2)

vectorized:

nums = np.array(nums)

result = nums * 2

Conceptually -

Loops ↓

element
↓
operation
↓
element
↓
operation

Numpy ↓

entire array
↓
optimized vectorized operation
↓
entire result

*Interview answer :

Vectorization means performing operations on entire arrays/matrices instead of manually iterating over elements in Python. Numpy executes these operations using optimized low-level routines, which makes numerical computations much faster and cleaner.

20. Broadcasting

Suppose -

$x = \text{np.array}([$
 $[1, 2, 3],$
 $[4, 5, 6]$
 $])$

$b = \text{np.array}([10, 20, 30])$

then

$x + b \implies$

Numpy roughly does :

1	2	3	+	10	20	30
4	5	6		10	20	30

result -

11 22 33

14 25 36

In neural networks,

$$Z = XW + b$$

b may have shape:

(10,)

while:

$$XW = (1000, 10)$$

Numpy broadcasts b across all 1000 samples.

21. Matrix Multiplication ***

A = np.array([
[1, 2],
[3, 4]
)

B = np.array([
[5, 6],
[7, 8]
)

$$C = A @ B$$

(or)

$$C = np.matmul(A, B)$$

Dimension rule:

$$(m \times n) @ (n \times p)$$

↓
(m × p)

example:

samples input features
↓ ↓
 $X = 1000 \times 20$
 $W = 20 \times 10$ ↓
 $XW = 1000 \times 10$ neurons

22. NumPy functions

np.zeros()
np.ones()
np.arange()
np.linspace()

np.sum()
np.dot()
np.matmul()

np.mean()
np.median()
np.std()
np.var()

np.reshape()
np.concatenate()
np.stack()

np.min()
np.max()
np.argmax()
np.random.rand()
np. " . randn()

ML formulas -

1. Standardization

$$z = \frac{x - \mu}{\sigma}$$

2. Euclidean distance

$$d = \sqrt{\sum (x_i - y_i)^2}$$

3. Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

4. ReLU

$$\max(0, x)$$

5. Softmax

$$\frac{e^{z_i}}{\sum_j e^{z_j}}$$

6. Mean

$$\mu = \frac{\sum x_i}{n}$$

7. Variance

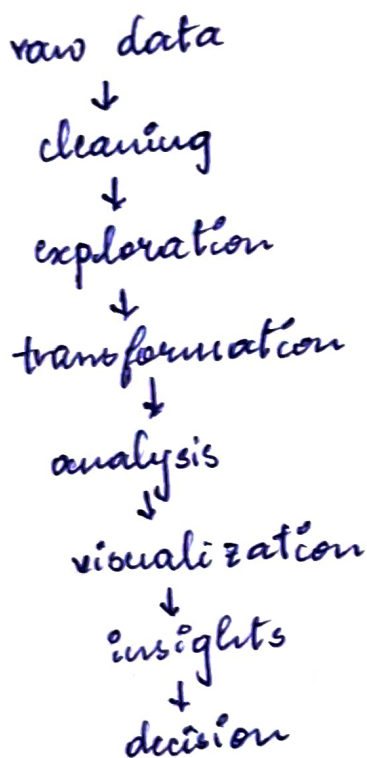
$$\text{Var}(x) = \frac{1}{n} \sum (x_i - \mu)^2$$

07/09/2026

DATA ANALYTICS + PANDAS

1. Data analytics - The process of examining, cleaning, transforming and interpreting data to discover useful patterns, answer questions and support decisions.

Basic flow :



2. Types

- i. Descriptive analytics : explains what has already happened using historical data.

example : What were our sales last month?

- ii. Diagnostic analytics : investigates why something happened by identifying relationships & causes.

example : Why did sales decrease last month?

iii. Predictive analytics: Uses historical patterns and statistical/machine-learning models to estimate what is likely to happen next.

example: What will next month's sales be?

iv. Prescriptive analytics: Recommends actions that should be taken to achieve the best outcome.

example: What pricing strategy should we use?

Descriptive → What happened?

Diagnostic → Why?

Predictive → What may happen?

Prescriptive → What should we do?

3. Data - A collection of observations, measurements, facts or values that can be analyzed to obtain information.

4. Dataset - An organized collection of related data, usually represented using rows and columns.

Age	Features	
	Salary	City
24	50000	Hyd
50	1200000	Pune
27	62000	Hyd

→ observations/records

5. Observation - A single record or row representing one entity or instance in a dataset.
6. Feature - An individual measurable input variable used to describe an observation.
examples: Age, salary, height, temperature.
7. Target variable - The output / dependent variable that we want to predict or analyze.
example: features $\rightarrow$ age, salary, experience
target $\rightarrow$ purchased product
8. Numerical Data - Data represented using measurable numerical values.

types
↓ ↓
discrete continuous

i. Discrete data :- Numerical data that contains countable values.

examples: no of students
" cars
" purchases

ii. Continuous data :- that can take any value within a range.

examples: height, weight, temp, salary.

9. Categorical Data - Data representing labels, groups or categories rather than measurable quantities.

examples: gender, city, department, color

i. Nominal data :- Categorical data whose categories have no meaningful order.

example: red, blue, green

ii. Ordinal data :- whose categories have a meaningful ranking / order

example: poor, average, good, excellent

10. Structured data - data stored using a predefined organized format such as rows and columns.

example: SQL tables, excel sheets, csv files

11. Unstructured data - data without a fixed tabular structure.

examples: images, videos, audio, docs, emails, etc..

12. Semi-structured data - data without a strict table structure but containing organizational tags/keys.

examples: JSON, XML

13. Data Cleaning: The process of detecting and correcting incomplete, inaccurate, duplicated or inconsistent data.

* typical pblms:

- i. missing values
- ii. duplicates
- iii. incorrect formats
- iv. invalid values
- v. outliers
- vi. inconsistent values

14. Missing values - Data entries where the expected value is unavailable / absent.

examples: NaN, NULL, None, blank cells.

* Common handling methods :-

i. remove rows

vi. replace with mode

ii. remove columns

vi. forward fill

iii. replace with mean

vii. backward fill

iv. replace with median

viii. imputation (model)

15. Mean imputation - replacing with arithmetic mean of available values.

* best used when :- 1. distribution $\approx$ symmetric
2. few outliers

16. Median imputation - replacing with the middle value of the distribution.

* useful when :- 1. outliers present 2. data is skewed.

17. Mode imputation - replace missing value with most frequently occurring values.

* usually useful for categorical values.

18. Duplicate data - two or more records containing the same / unnecessarily repeated information.

↳ can distort counts, avg, analysis, model training

19. Outlier - An observation that lies unusually far away from most other values in a dataset.

example: Salary - 35k 40k 42k 45k 900k

↑
outlier

*** An outlier is not automatically an error.

It may represent:

- data-entry mistake
- rare event
- fraud

20. IQR (Interquartile range) - The diff b/w 3rd and 1st quartile and is commonly used to measure data spread and detect outliers.

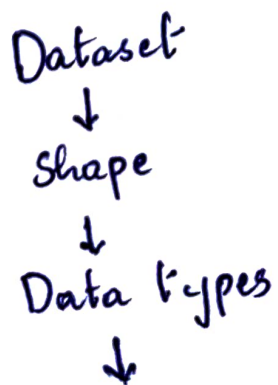
$$IQR = Q_3 - Q_1$$

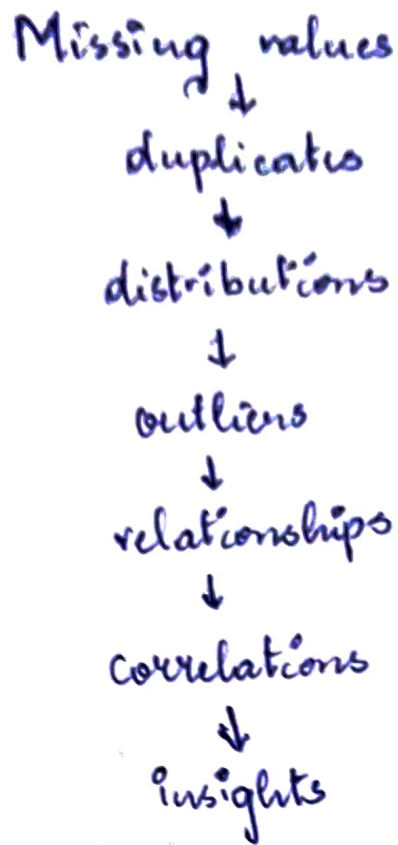
$$\text{lower} = Q_1 - 1.5(IQR)$$

$$\text{upper} = Q_3 + 1.5(IQR)$$

21. Exploratory data analysis (EDA) - The process of exploring a dataset using statistics and visualizations before building models / drawing conclusions.

Flow -





22. Descriptive statistics -

Statistical measures used to summarize and describe the main characteristics of a dataset.
common measures:

mean, median, mode, minimum, maximum,
range, variance, standard deviation, outliers

23. Mean - The arithmetic average of a set of values.

$$= \frac{\sum x_i}{n} = \frac{\sum x}{n}$$

eg: 2, 4, 6 $\Rightarrow$ mean = 4

24. Median - The middle value after arranging values in sorted order. (less sensitive to outliers than mean)

eg: 1, 4, 8, 10, 50
↓
median

25. Mode - The value occurring most frequently.
eg: 1 2 2 3
mode = 2

26. Range - The diff btw max and min values
range = max - min

27. Variance - measures how far data values are spread from their mean.

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n}$$

higher variance :- values are widely spread

lower variance :- values remain closer to mean

28. Standard deviation - The square root of variance and represents the typical spread of values around the mean.

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$$

29. Percentile - indicates the % of observations that fall below a particular value.

eg: 90th percentile means roughly -

90% of observations are below that value.

30. Quartile - A value dividing sorted data into 4 equal portions.

$Q_1 = 25^{\text{th}}$ percentile

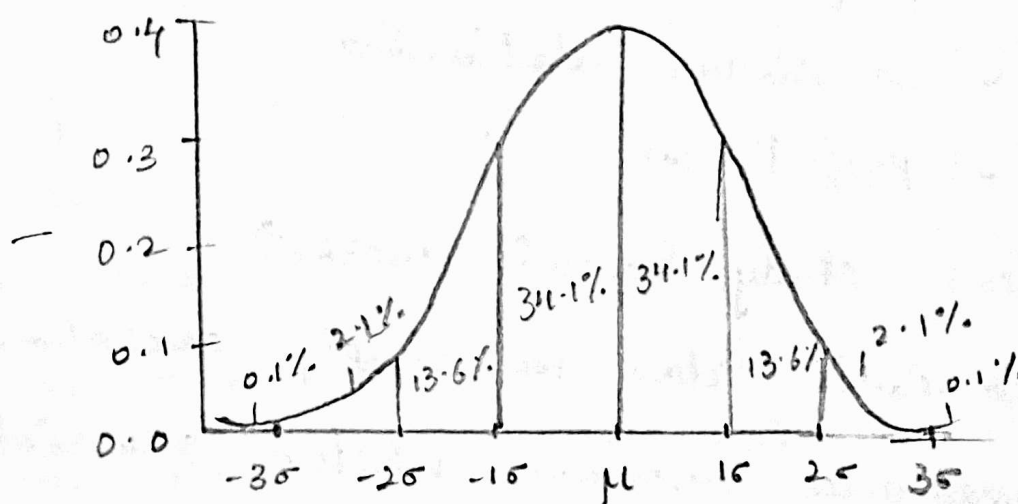
$Q_3 = 75^{\text{th}}$ percentile

$Q_2 = 50^{\text{th}}$ percentile = median

31. Distribution -

describes how values of a variable are spread and how frequently different values occur.

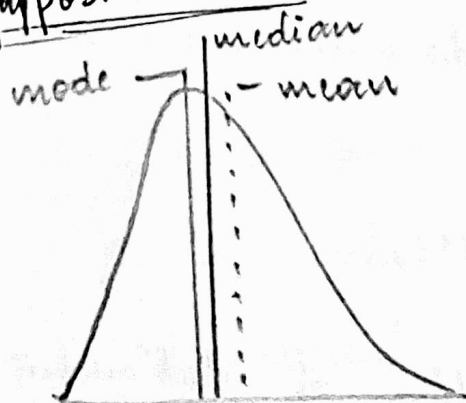
32. Normal distribution - A symmetric bell-shaped probability distribution where values tend to cluster around the mean.



mean $\approx$ median $\approx$ mode

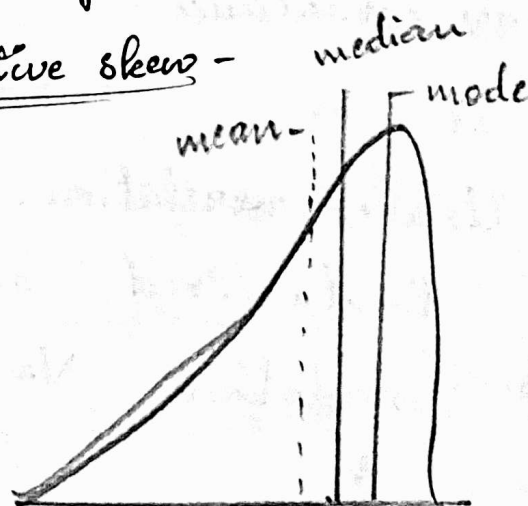
33. Skewness - measures asymmetry of a distribution.

right/positive skew -



mean $>$ median

left/negative skew -



mean $<$ median

right skew - long tail towards larger values.

left skew - long tail towards smaller values.

34. Correlation - measures the strength and direction of the relationship b/w 2 variables.

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

+1 $\rightarrow$ perfect +ve relationship

0 $\rightarrow$ no linear relationship

-1 $\rightarrow$ perfect -ve "

examples: study hours $\uparrow$ marks $\uparrow$

** correlation does not imply causation

35. Covariance - measures whether 2 variables tend to increase / decrease together.

+ve covariance: $\rightarrow$ move together.

-ve " : $\rightarrow$ move oppositely

* Unlike correlation, covariance does not have a standardized range.

36. Correlation Vs Covariance

$\downarrow$
direction +

standardized strength

$\downarrow$
direction of relationship

37. Data visualization - The graphical representation of data to make patterns, trends, relationships, and anomalies easier to understand.

38. Bar chart: compares numerical values across different categories.

↳ sales by city, students by department

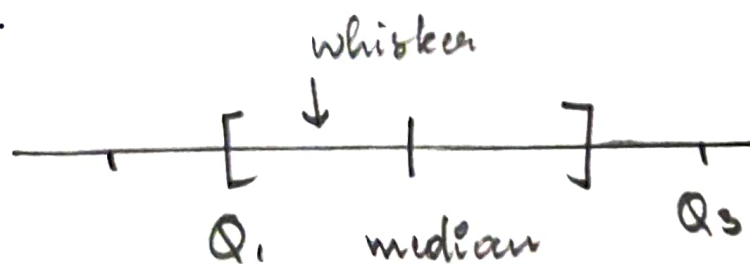
39. Line chart: displays how a numerical var changes across an ordered sequence. (time)

↳ monthly sales, stocks; temperature

40. Histogram: displays the freq distribution of a continuous numerical variable using intervals called bins.

↳ distribution, skewness, spread

41. Box plot: Summarizes a numerical distribution using quartiles, median, spread and potential outliers.



possible dots outside whiskers = outliers

42. Scatter plot: shows the relationship b/w 2 numerical values using individual data points.

* useful for: clusters, outliers, relationships, correlation

43. Heatmaps: Represents numerical values using variations in color intensity.

* common ML use - correlation matrix.

44. Feature Scaling - transforms numerical features so that variables measured on different scales become comparable.

example: age $\rightarrow$ 25

salary $\rightarrow$ 9000000

without scaling, algorithms using distance/gradients may be disproportionately influenced by salary.

45. Standardization - rescales a features so that it has approximately zero mean and unit standard deviation.

$$Z = \frac{x - \mu}{\sigma}$$

$\nearrow$ data value
 $\rightarrow$ mean
 $\rightarrow$ sd

after standardization:

mean ≈ 0 sd ≈ 1

commonly used for -

- logistic regression
- SVM
- KNN
- PCA
- Neural networks

46. Normalization - rescales values into a specified range, commonly 0 to 1.

Min - Max :

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

typical range :- $0 \rightarrow 1$.

47. Standardization $\forall$ Normalization
 $\downarrow$ $\downarrow$
mean $\rightarrow 0$ usually maps values
std $\rightarrow 1$ between 0 and 1

does not necessarily bound values.

48. Categorical encoding - converts categorical values/variables into numerical representations that machine-learning algorithms can process.

49. Label encoding - assigns an integer value to each category.

eg: low $\rightarrow 0$

medium $\rightarrow 1$

high $\rightarrow 2$

50. One - Hot encoding - creates separate binary columns for each categorical value.

eg: Color $\rightarrow$ red, blue, green

becomes :	red	blue	green
	1	0	0
	0	1	0
	0	0	1

useful when categories have no natural order.

51. Feature Engineering -

The process of creating, selecting, modifying or combining features to improve analysis / model performance.

eg :

DOB
↓
calculate
↓
age

price x quantity
↓
total purchase

52. Feature Selection - The process of choosing the most relevant input var while removing redundantⁿ features.

eg : Benefits →

less overfitting.

faster models.

simpler interpretation.

lower computation.

53. Data leakage - Occurs when information unavailable during real prediction accidentally

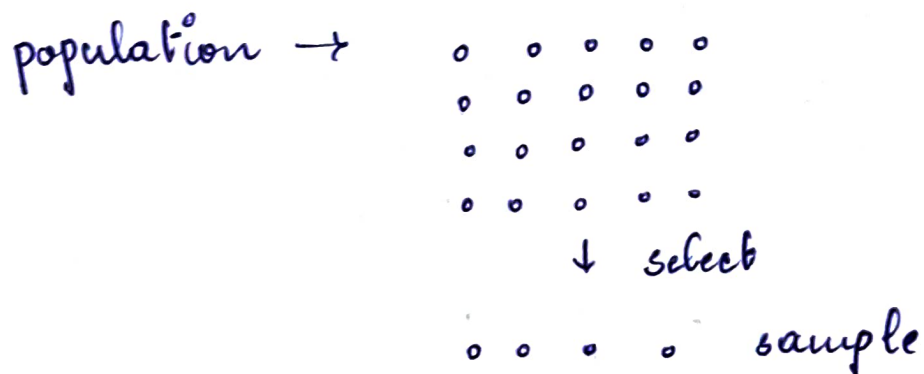
enters / model training and produces unrealistically good performance. / unintentional exposure of private info
example :

predicting whether someone receives a loan while including :
loan approved data → input feature.

54. Sampling - The process of selecting a smaller subset of observations from a population for analysis.

55. Population - The complete set of individuals, obj or observations we want to study.

56. Sample - A subset of population used to estimate characteristics of the full population.



57. Random Sampling - method where every member of the population has an approximately equal chance of being selected.

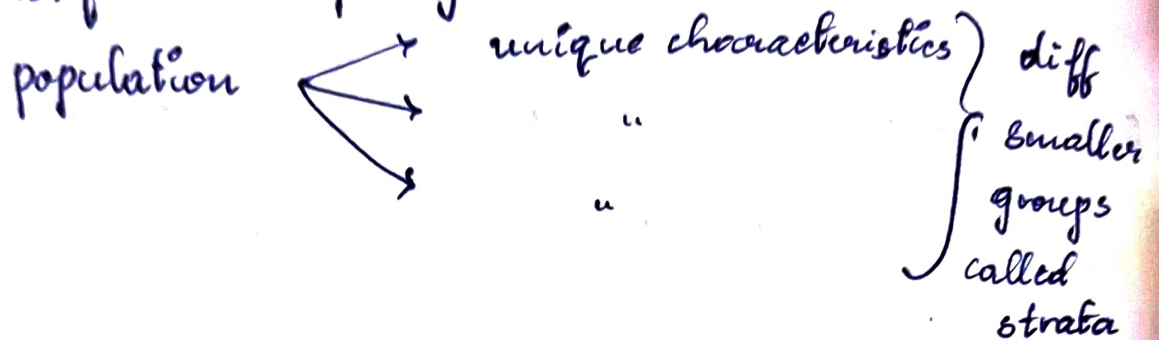
58. Bias - A systematic error that causes measurements, samples / models to consistently favor certain outcomes.

Types of probability sampling -

i. Simple random sampling:

equal chance of getting selected.

ii. Stratified sampling:



also known as mixed sampling

purposive random

iii. Systematic sampling:

arranged in specific order, every nth item is selected.

example: take every 20th bulb off.

↓

tests for defects.

iv. Cluster sampling: (research method)

split large population into groups

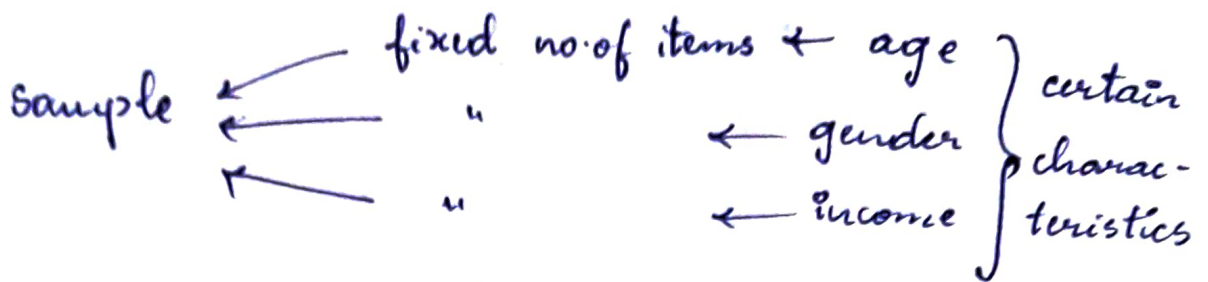
↓

randomly pick a few of these (schools, neighborhoods)

groups and study everyone

Types of non-probability sampling -

i. Quota sampling :



ii. Convenience sampling :

→ investigator selects items based on convenience.

iii. Purposive / Deliberate sampling :

→ selects items that are most relevant to research objectives.

iv. Snowball sampling : (research method)

→ sample is increased when existing subjects recruit by referrals.

59. Pandas -

A python library designed for manipulating, cleaning, transforming and analyzing structured tabular data.

→ import pandas as pd.

60. Series - one-dimensional labeled array

Index	Value
0	Apple
1	Mango
2	Orange

61. Dataframe - A 2D labeled table containing rows and columns.

	Name	Age	marks
0	A	20	85
1	B	21	92
2	C	22	78

62. Reading a csv -

`read_csv()` : loads CSV data into pd DF.

→ `df = pd.read_csv("data.csv")`

63. head() - displays 1st few rows of a DataFrame.

→ `df.head()`

default : 1st 5 rows.

64. tail() - displays last few rows

→ `df.tail()`

65. shape - returns the no. of rows and columns.

→ `df.shape`

%p - (1000, 12)
 ↓ ↓
 rows cols

66. columns - returns the column labels.

→ `df.columns`

67. dtypes - shows datatypes stored in each DF column
common Pandas - types:

- int64
- float64
- object
- bool
- datetime64

68. info() - provides a concise summary of a DF including columns, datatypes, and non-null counts.

→ `df.info()`

69. describe() - generates descriptive statistical information for numerical columns.

→ count, mean, std, min, 25%, 50%, 75%,
max

70. selecting a column -

→ `df["Age"]`

• returns 1 column
usually - series.

71. selecting multiple col -

→ `df[["Age", "Salary"]]`

usually returns: DataFrame

72. loc - selects rows and cols using labels.

→ `df.loc[row-label, col-label]`

example: `df.loc[0, "Age"]`

73. iloc - select rows and cols using integer positions.

→ `df.iloc [row-position, column-position]`

example: `df.iloc [0, 2]`

74. filtering - selecting rows that satisfy a specified condition.

→ `df [df ["Age"] > 25]`

multiple conditions:

→ `df [(df ["Age"] > 25) &
(df ["Salary"] > 50000)]`

75. isnull() - identifies missing values in a DF.

→ `df.isnull()`

count: → `df.isnull().sum()`

76. dropna() - removes rows/cols containing missing values.

→ `df.dropna()`

77. fillna() - replaces missing values using a specified value / strategy.

→ `df ["Age"].fillna(df ["Age"].mean())`

78. duplicated() - identifies repeated rows.

→ `df.duplicated()`

79. drop_duplicates() - removes duplicate rows.

→ df.drop_duplicates()

80. value_counts - counts how freq each unique value occurs.

→ df["City"].value_counts()

* excellent for inspecting categorical values.

81. unique() - returns the distinct values contained in a column.

→ df["City"].unique()

82. nunique() - returns no. of unique values

→ df["City"].nunique()

83. sort_values() - sorts DF rows according to one or more columns.

→ df.sort_values("Salary")

descending: df.sort_values(
"Salary",
ascending = False
)

84. groupby() - groups rows according to common categorical values so aggregate calculations can be performed.

→ df.groupby("Department")["Salary"].mean()

employees
↓
group by department
↓
HR IT Sales
↓
calculate avg salary

85. Aggregation - combines multiple values into a summary statistic.

examples: • sum • count • max
• mean • min • median

```
→ df.groupby("Department").agg({  
    "Salary": "mean",  
    "Age": "median"  
})
```

86. Merge - combines DataFrames using a common key, similar to SQL JOIN.

```
→ pd.merge(df1, df2, on="ID")
```

* common joins:

• inner • right
• left • outer

87. Inner Join - keeps only matching rows in both datasets.

→ $A \cap B$

88. Left Join - keeps every row from the left dataset and matching ones from right dataset.
89. Right Join - keeps every row from right dataset and matching ones from left dataset.
90. Outer Join - keeps all rows from both datasets whether they match or not.
91. concat() - combines pandas objects along rows or columns.
→ `pd.concat([df1, df2])`

92. Pivot table - Summarizes and recognizes data by grouping categories into rows and columns with aggregated values.

example :

	product	
	A	B
City		
Hyd	50	30
Pune	80	20

93. apply() - applies a custom function to values along a Series / DataFrame axis.
→ `df["Age"].apply(lambda x: x**2)`

94. Datetime - A specialized datatype used for storing and manipulating dates and times.

• conversion : `pd.to_datetime(df["Date"])`

* useful $\rightarrow$ `df["Date"].dt.year`
`df["Date"].dt.month`
`df["Date"].dt.day`

95. Correlation Matrix -

A table showing pairwise correlation coefficients between numerical variables.

$\rightarrow$ `df.corr(numeric_only=True)`

	Age	Salary	Experience
Age	1	.45	.72
Salary	.45	1	.81
Exp	.72	.81	1

1 because every var perfectly correlates with itself.

96. End-to-End Data Analytics flow

BUSINESS PROBLEM



COLLECT DATA



UNDERSTAND DATA



